

Manendra Singh

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PROFESSIONAL SUMMARY

Computer Science undergraduate with strong problem-solving skills and hands-on experience in software development, data structures & algorithms, machine learning, and full-stack application development. Solved 400+ LeetCode and 350+ GeeksforGeeks problems, demonstrating strong analytical and coding abilities. Experienced in Java, Python, SQL, JavaScript, React, Node.js, and MySQL through academic and personal projects. Passionate about building scalable, real-world solutions, learning emerging technologies, and contributing effectively in collaborative, fast-paced environments.

EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu <i>Bachelor of Technology in Computer Science and Engineering</i>	2023 – 2027 CGPA: 8.52
Swami Keshwanand Shikshan Sansthan, Sikar, Rajasthan <i>Higher Secondary School (12th Standard)</i>	2021 – 2022 Percentage: 91.4
Swami Keshwanand Shikshan Sansthan, Sikar, Rajasthan <i>High School (10th Standard)</i>	2019 – 2020 Percentage: 88.33

PERSONAL PROJECTS

Diabetic Retinopathy Detection – Deep Learning + Grad-CAM (Live Demo) <i>A clinical AI web app for automated detection and classification of diabetic retinopathy stages from retinal fundus images.</i> • Tech Stack: Python, TensorFlow, EfficientNet, ResNet, OpenCV, Grad-CAM, Streamlit, FPDF • Trained deep learning models (EfficientNet, ResNet, custom CNN) using transfer learning with data augmentation and class imbalance handling to classify DR severity into 5 stages (Healthy to Severe). • Integrated Grad-CAM heatmap visualization to highlight pathological retinal regions, improving model interpretability and clinical trustworthiness. • Built a Streamlit dashboard supporting multi-image batch processing and automated PDF report generation with patient details, confidence scores, and Grad-CAM overlays.	GitHub
SCSMS – Smart City Service Management System (Live Demo) <i>A real-time smart city platform for streamlined service reporting and citizen-admin communication.</i> • Tech Stack: Node.js, Express, MySQL (Aiven), Socket.IO, Leaflet.js, OpenStreetMap API, JavaScript • Built a scalable and secure backend system enabling real-time communication, reducing issue reporting latency by 40% using Socket.IO. • Designed a secure relational database schema with role-based access control (RBAC), indexing, and normalized table structures to ensure data integrity and efficient query resolution. • Developed RESTful APIs following MVC architecture, ensuring maintainability, modularity, and adherence to software engineering best practices. • Collaborated on a modular, team-friendly codebase with clear separation of concerns, enabling parallel development and easier onboarding.	GitHub
Stack-IT (Live Demo) <i>A community-driven discussion forum for developers to share technical knowledge through questions and answers.</i> • Tech Stack: MongoDB, Mongoose, Express.js, Node.js, JavaScript, HTML5, CSS3 • Designed and deployed a scalable platform with secure authentication and authorization mechanisms (JWT, bcrypt). • Engineered optimized database schemas with indexing strategies and query tuning to improve data retrieval speed and overall platform scalability. • Developed a dynamic ranking algorithm based on user votes and engagement metrics to improve answer relevance and content discoverability.	GitHub

TECHNICAL SKILLS

Programming Languages: Java, C, C++, Python, SQL, JavaScript
Web Tech: Node.js, Express.js, HTML, CSS
Databases & Cloud: MongoDB, MySQL, Firebase (Cloud-based DB)
Core CS Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems
Software Engineering: REST APIs, MVC Architecture, Scalable Systems, Secure Coding, Debugging
Developer Tools: Git, GitHub, Visual Studio Code

CERTIFICATIONS / ACHIEVEMENTS

LeetCode: Resolved 400+ problems Peak Rating: 1743 Global Rank < 3,00,000	Profile
GeeksforGeeks: Resolved 350+ problems Global Rank: 30,000 Institute Rank < 500	Profile
Complete Machine Learning & Data Science – Skill Up	GeeksforGeeks / Nation SkillUp
AI Agent Workshop using AgentForce	GeeksforGeeks & Salesforce – July 2025
Attended Google Cloud Agent AI Day	Hack2Skill – July 2025
Competed in Adobe India Hackathon	Adobe – July 2025